

Artificial Intelligence (Part 1a)

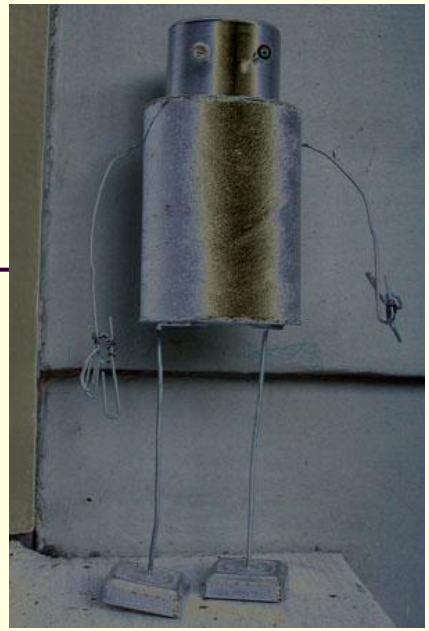
Overview of AI history

Course Contents

Again..Selected topics for our course. Covering all of AI is impossible!
Key topics include:

- Introduction to Artificial Intelligence (AI)
- Knowledge Representation and Search
- Introduction to AI Programming
- Problem Solving Using Search
- Exhaustive Search Algorithm
- Heuristic Search
- Techniques and Mechanisms of Search Algorithm
- Knowledge Representation Issues and Concepts
- Strong Method Problem Solving
- Reasoning in Uncertain Situations
- Soft Computing and Machine Learning

What is intelligence?



the ability to comprehend, to understand and profit from experience ??

For thousands of years people tried to understand *how we think*

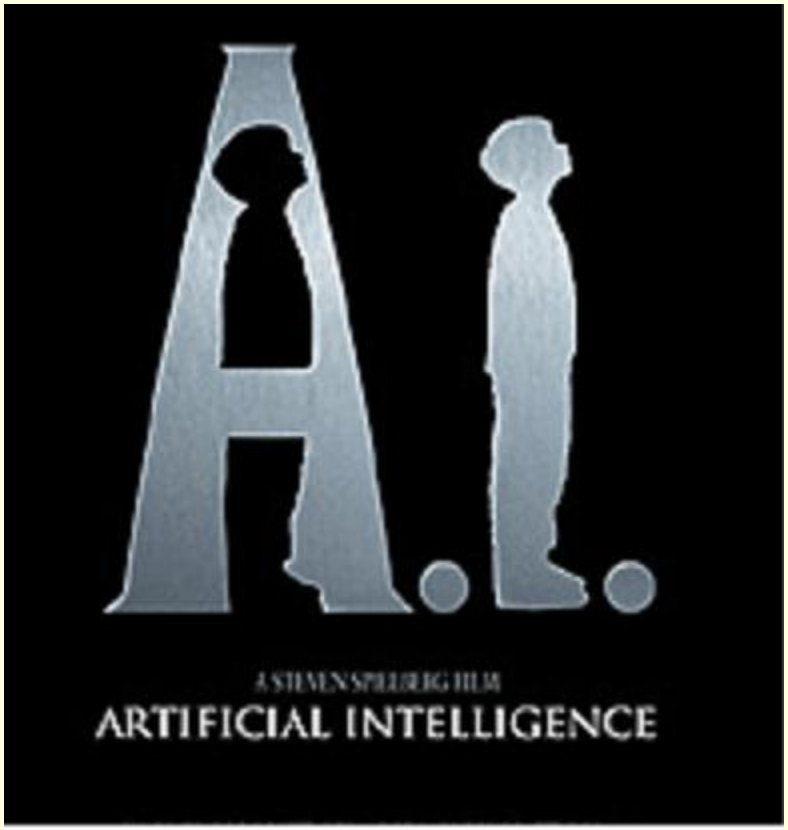
- Philosophy
- Mathematics
  - What is correct mathematical reasoning?
- Neuroscience
- Psychology
- Economics

Oral Quiz

- Ayah Magi mempunyai 5 orang anak,
  - Nana
  - Nini
  - Nene
  - Nono.. Siapakah nama anak kelimanya?



50 years of Artificial Intelligence



A Bit of History

- ENIAC: first super-computer, created in 1946
- Summer 1956: “Dartmouth Summer Research Project on Artificial Intelligence”
- John McCarthy (also known for LISP, time-sharing) coined the term “artificial intelligence”
- Organizers: Marvin Minsky (co-founder with McCarthy of MIT AI lab, pioneer and detractor of neural nets), Nathaniel Rochester (IBM), and Claude Shannon (Bell Labs, inventor of information theory, also had a robot that did open-loop 5-ball juggling)
- Herbert Simon (Nobel prize in economics) and Allen Newell presented “Logic theorist” (an automated theorem prover), one of the first operating programs in AI

Early AI Hopes and Dreams

- Make programs that exhibit similar signs of intelligence as people: prove theorems, play chess, have a conversation
- Learning from experience was considered important
- Logical reasoning was key
- The research agenda was geared towards building “general problem solvers”
- There was a lot of hope that natural language could be easily understood and processed

50 years later: DARPA Grand Challenge

- DARPA (agency), offered a $2 million prize for US military research funding an automated driving competition (autonomous, driverless)
- Task: Drive through the Nevada desert 132 miles, start and finish specified the morning of the competition, *with no input from any human*
- 2004 competition, none finished, the best robot car (CMU Red Team) crashed after 7.36 miles
- 2005 competition, *5 robots* (out of 23) finished the race
- The winning robot, Stanley (from Stanford Univ.) finished in 6 hours 54 minutes.

From NY Times

The five robots that successfully navigated a 132-mile course in the Nevada desert last weekend demonstrated the re-emergence of artificial intelligence [...] The winning robot, named Stanley, covered the unpaved course in just 6 hours and 53 minutes without human intervention and guided only by global positioning satellite(GPS) waypoints. The feat, which won a $2 million prize from the Pentagon Defense Advanced Research Project Agency, was compared by exuberant Darpa officials to the Wright brothers' accomplishment at Kitty Hawk, because it was clear that it was not a fluke. [...] The ability of the vehicles to complete a complex everyday task - driving - underscores how artificial intelligence may at last be moving beyond the research laboratory.

Stanley

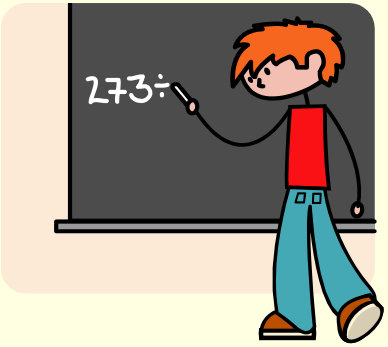
- Built based on a Volkswagen car
- An array of sensors: cameras, laser range finders, radar, GPS
- *Probabilistic reasoning* and *machine learning* algorithms are the heart of the software
- The robot is capable of assessing how good the data is, based on prior training

What is AI?

Quick Answers from the Media

- What socially inept super-hackers do
- The opposite of natural stupidity
- Data
- Deep Blue, Sony dancing robots, poker players





Quick Answers from Academia

- Modeling human cognition using computers
- The study of problems that other CS folks do not yet know how to do
- **Cool stuff!!!**

  Game playing agents, machine learning, data mining, speech, language, vision, web agents, chatbots, robots, ...
- **Useful stuff!!!**

  Medical diagnosis, fraud detection, genome analysis, object identification, space shuttle scheduling, information retrieval, ...

What should AI Systems Do?

What is Artificial Intelligence?

A mix of general purpose and special purpose algorithms

- General purpose
  - Learning algorithms
  - Theorem proving
- Special purpose
  - Recognizing visual patterns
  - Language modelling

All are integrated not quite so seamlessly...

Goals of AI

Systems that think like humans	**Systems that think rationally**
Systems that act like humans	**Systems that act rationally**

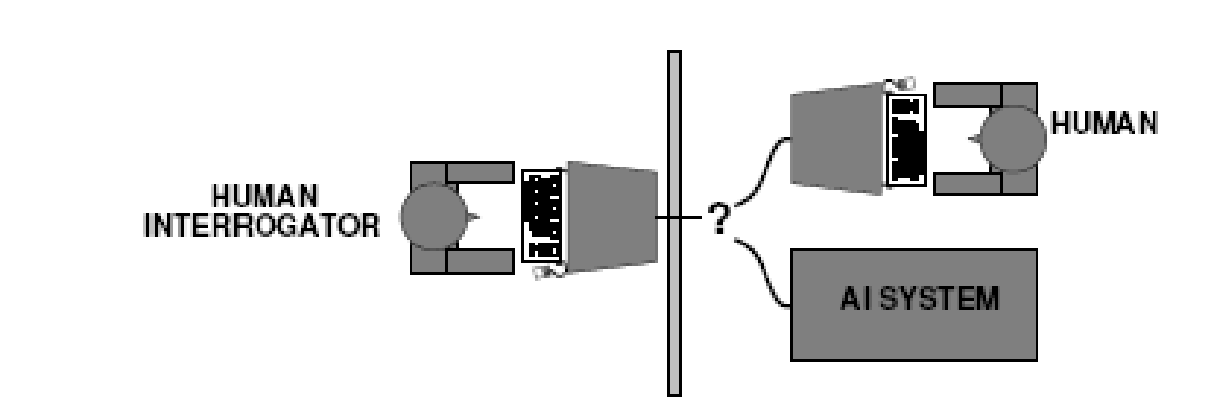
- AI as *acting humanly* --- as typified by the Turing test
- AI as *thinking humanly* --- cognitive science.
- AI as *thinking rationally* --- as typified by logical approaches.
- AI as *acting rationally* --- the intelligent agent approach.

Acting Humanly

- Emphasis on how to *tell* that a machine is intelligent, not on how to *make it* intelligent, *when can we count a machine as being intelligent*?

"Can machines think?" → **"Can machines behave intelligently?"**

- Most famous response due to Alan Turing, British mathematician and computing pioneer:



Turing test

- No program has yet passed Turing test!
  (Annual Loebner competition & prize.)
- A program that succeeded would need to be capable of:
  - natural language understanding & generation;
  - knowledge representation;
  - learning;
  - automated reasoning.
- Note no *visual* or *aural* component to basic Turing test

-- augmented test involves video & audio feed to entity.

Problem: Turing test is not *reproducible*, *constructive*, or amenable to *mathematical analysis*

Thinking Humanly

- Try to understand how the mind works; how do we think?
- Two possible routes to find answers:
  - by *introspection* -- we figure it out ourselves!
  - by *experiment* -- draw upon techniques of psychology to conduct controlled experiments. (“Rat in a box”!)
- The discipline of *cognitive science*: particularly influential in *vision*, *natural language processing*, and *learning*.

Human vs Machine Thinking (I)

Expert systems -- AI success story in early 80's.

- Human expert's knowledge and experience is passed to a computer program
- Rule-based representation of knowledge
- Typical domains are:
  - medicine (INTERNIST, MYCIN, . . .)
  - geology (PROSPECTOR)
  - chemical analysis (DENDRAL)
  - configuration of computers (R1)

Thinking humanly works

Human vs Machine Thinking (II)

Computer program playing chess

- "Human way"
  - Tried by World champion M.Botvinnik (who also was a programmer)
  - Poor performance
- "Computer way"
  - Sophisticated search algorithms
  - Vast databases
  - Immense computing power
  - Human world champion beaten

Thinking Rationally

- Trying to understand how we *actually* think is one route to AI -- but how about how we *should* think.
- Use *logic* to capture the *laws of rational thought* as *symbols*.
- *Reasoning* involves shifting symbols according to well-defined rules (like algebra).
- Result is *idealised* reasoning.

Logic and AI

- Logicist approach theoretically attractive.
- Lots of problems:
  - *transduction* -- how to map the environment to symbolic representation;
  - *representation* -- how to represent real world phenomena (time, space, . . .) symbolically;
  - *reasoning* -- how to do symbolic manipulation *tractably* -- so it can be done by real computers!

Acting Rationally (I)

- Acting rationally = acting to achieve one's goals, given one's beliefs.
- An *agent* is a system that perceives and acts; intelligent agent is one that acts rationally w.r.t. the goals we delegate to it.
- Emphasis shifts from designing *theoretically best* decision making procedure to best decision making procedure possible in *circumstances* (reasoning*)*.
- Logic may be used in the service of finding the best action -- not an end in itself.

Acting Rationally (II)

- Achieving *perfect* rationality -- making the best decision theoretically possible -- is not usually possible, due to *limited resources*:
  - limited time;
  - limited computational power;
  - limited memory;
  - limited or uncertain information about environment.
- The trick is to *do the best with what you've got*!

Example AI System: Chess Playing

E.g. Deep Blue (IBM)

- Perception: advanced features of the board
- Actions: choose a move
- Reasoning: heuristics to evaluate board positions, search

Example AI System: Medical Diagnosis

E.g. Pathfinder (D.Heckerman, Microsoft Research)

- Perception: symptoms, test results
- Actions: suggest tests, make diagnosis
- Reasoning: Bayesian inference, machine learning, Monte Carlo simulation

Example AI System: Automatic Driver

E.g. ALVINN (D.Pomerleau, CMU)

- Perception: digitized camera image of the road
- Actions: 64 different steering angles
- Reasoning: artificial neural network trained with backpropagation

Example AI System: Scheduling

- Perception: task description language
- Action: trivial
- Reasoning: constraint satisfaction, linear programming, genetic algorithms

Classical AI

- Reasoning was once seen as ***the*** AI problem
- Chess (and related games) were considered pivotal to understanding intelligence
- Golden goal: a *general problem solver*
- But general problem solving is hard! Memory and computational complexity issues hit in early 70s
- Instead, research moved to rule-based expert systems in the 80s, and discovered their limitations too...

How should we represent knowledge? Is logic enough?

Recent AI: Math to the Rescue!

- Heavy use of probability theory, decision theory, statistics
- Trying to solve *specific problems* rather than build a general reasoner
- AI today is a collection of sub-fields:
  - Perception (especially vision) is a separate community
  - Robotics is also largely separate
  - Deliberative reasoning is now the part named “AI”
  - Even within reasoning, different approaches evolved, with different styles and terminology
- A lot of progress was made in this way!

State of the Art

Which of the following can be done at present?

- Play a decent game of table tennis
- Play a decent game of bridge
- Discover and prove a new mathematical theorem
- Write an intentionally funny story
- Give competent legal advice in a specialized area of law
- Translate spoken English into spoken Swedish in real time

Oral Quiz…

- The more you learn, the more you realise you know _____ and _____

SO…

STOP LEARNING *&^%$#?

